

DAMG7370 - FINAL REPORT

APRIL 14, 2025

DATA ARCHITECTURE DESIGN FOR INSTACART DATASET GROUP 1

Team contribution

NUID	Name	Role	Responsibilities
002643735	Sujiitra Murukeshan	Architecture Lead	• Designed end-to-end architecture using Medallion principles • Outlined data flow across Azure services • Created architecture diagrams
002651640	Thamashi Kumarasingh e	Data Pipeline Engineer	• Uploaded dataset from Kaggle to Azure Blob Storage • Built dynamic pipelines in Azure Data Factory • Ingested data into Data Lake Gen2
002832662	Tejas Samudrala	Data Processing Engineer	• Wrote PySpark transformation scripts in Databricks • Cleaned and cast datasets • Saved results to Silver zone in Parquet/Delta format
002700327	Aishwarya Sreenivasan	BI & Reporting Engineer	• Created external data source & SQL views in Synapse • Connected Synapse to Power BI • Designed and published dashboard visualizations

1. Introduction

Instacart is a grocery delivery and pickup service that connects customers with personal shoppers. It contains over 3 million grocery orders placed by more than 200,000 users. It offers a rich source of transactional data suitable for building recommender systems and understanding consumer behaviour. This dataset consists of several CSV files, such as orders, order_products__prior.csv & order_products__train.csv, products, aisles and departments.

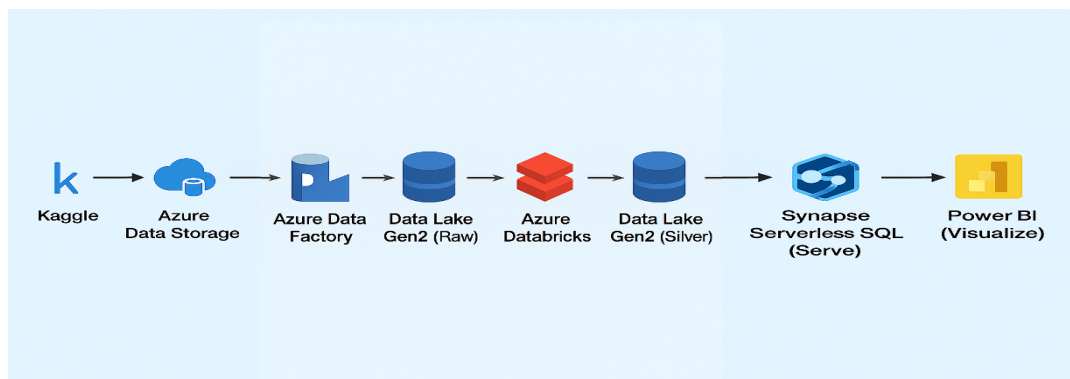
This project involved designing and implementing an end-to-end cloud-based data architecture pipeline using the Microsoft Azure ecosystem. We aimed to take a large real-world dataset and apply modern data engineering practices to clean, transform, and analyze it efficiently.”

2. Objective

The primary objectives of the project were:

- To simulate a real-world enterprise-level data pipeline using Azure tools.
- To design a modular and scalable architecture.
- To clean and transform data using PySpark on Azure Databricks.
- To expose transformed data for analytics via Synapse Serverless SQL.
- To build interactive visualizations using Power BI.

3. Data Architecture



This end-to-end data pipeline is built on **Microsoft Azure services**, enabling seamless data ingestion, transformation, storage, and visualization.

1. Kaggle (Source):

The dataset is sourced from Kaggle and serves as the raw data input for the pipeline.

2. Azure Data Storage:

The data is initially stored in Azure Blob Storage, providing a scalable and secure foundation.

3. **Azure Data Factory:**
This orchestrates data movement from Blob Storage into **Azure Data Lake Gen2 (Raw)** using ETL processes.
4. **Data Lake Gen2 (Raw):**
Acts as the **Bronze Layer**, storing unprocessed data for further refinement.
5. **Azure Databricks:**
Utilized for data cleaning, transformation, and enrichment using Spark-based notebooks.
6. **Data Lake Gen2 (Silver):**
Stores the transformed data in a structured format (the **Silver Layer**) ready for analytical querying.
7. **Synapse Serverless SQL:**
Enables querying and serving of curated data from the Silver Layer without the need for infrastructure provisioning.
8. **Power BI:**
Finally, the processed data is visualized in Power BI, enabling interactive dashboards and business insights.
9. **GitHub:** For storing code artifacts, notebooks, and supporting documentation.

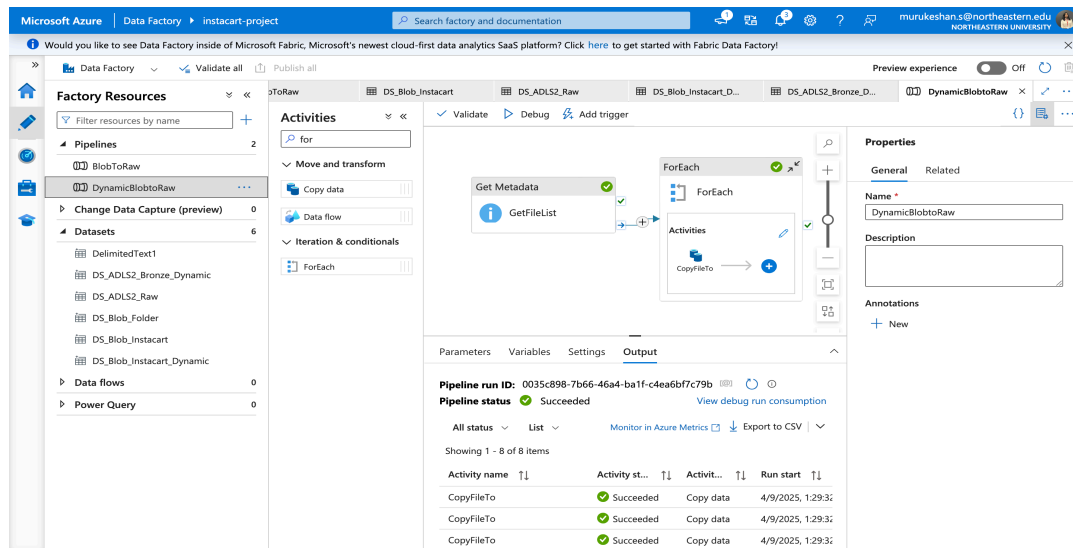
We adopted the Medallion Architecture:

- Bronze Layer: Raw ingested data from Blob Storage.
- Silver Layer: Cleaned and transformed data stored in Delta/Parquet format.
- Gold Layer: Logical layer consisting of Synapse SQL views for analysis.

4. Pipeline

DynamicBlobtoRaw pipeline was executed within Azure Data Factory (ADF), part of the "instacart-project". The pipeline is designed to dynamically ingest files from an Azure Blob Storage container and move them into the Raw layer of Azure Data Lake Storage Gen2 (ADLS Gen2). The process begins with the Get Metadata activity (GetFileList), which fetches a list of files from the source storage. This list is then passed into a ForEach loop, allowing the pipeline to iterate through each file. Inside the loop, the CopyFileTo activity is triggered to perform the actual data copy from Blob Storage to the raw zone of ADLS Gen2.

The Output panel confirms that the pipeline run was successful, with all eight activities completing without errors. Each CopyFileTo task is marked as Succeeded, indicating that multiple files were effectively transferred on April 9, 2025, at 12:39 PM. The left panel lists associated datasets such as DS_Blob_Instacart (source) and DS_ADLS2_Raw (sink), showcasing the integration of reusable components for flexible data movement. This dynamic pipeline setup is an efficient and automated solution for batch ingestion of files, making it ideal for projects that deal with continuously updating datasets like those from Instacart.



5. Data Ingestion Layer

The data ingestion layer is the foundation of the pipeline, responsible for bringing raw data into the system. In this project, the process begins with creating a resource group, followed by provisioning a Storage Account and containers in Azure Blob Storage. This serves as the initial landing zone for the datasets.

Once the storage environment is prepared, Azure Data Factory (ADF) is used to automate the ingestion process. A pipeline named `DynamicBlobtoRaw` is created to dynamically load files from the Blob container into the Raw zone of Azure Data Lake Storage Gen2. The pipeline consists of key activities like:

- **Get Metadata:** Collects a list of files available in the Blob container.
- **ForEach:** Iterates over each file to copy it.
- **Copy Data:** Transfers each file to the Data Lake Gen2 (Bronze layer), preserving its raw state.

This design ensures the ingestion process is both dynamic and scalable, allowing for easy addition of new data without manual intervention.

Name	Modified	Access tier	Archive status	Blob type	Size	Lease state
[.]						...
aisles.csv	4/9/2025, 1:29:44 AM	Hot (Inferred)		Block blob	2.54 KiB	Available
departments.csv	4/9/2025, 1:29:42 AM	Hot (Inferred)		Block blob	270 B	Available
order_products_prior.csv	4/9/2025, 1:29:46 AM	Hot (Inferred)		Block blob	550.8 MiB	Available
order_products_train.csv	4/9/2025, 1:29:42 AM	Hot (Inferred)		Block blob	23.54 MiB	Available
orders.csv	4/9/2025, 1:29:45 AM	Hot (Inferred)		Block blob	103.92 MiB	Available
products.csv	4/9/2025, 1:29:44 AM	Hot (Inferred)		Block blob	2.07 MiB	Available

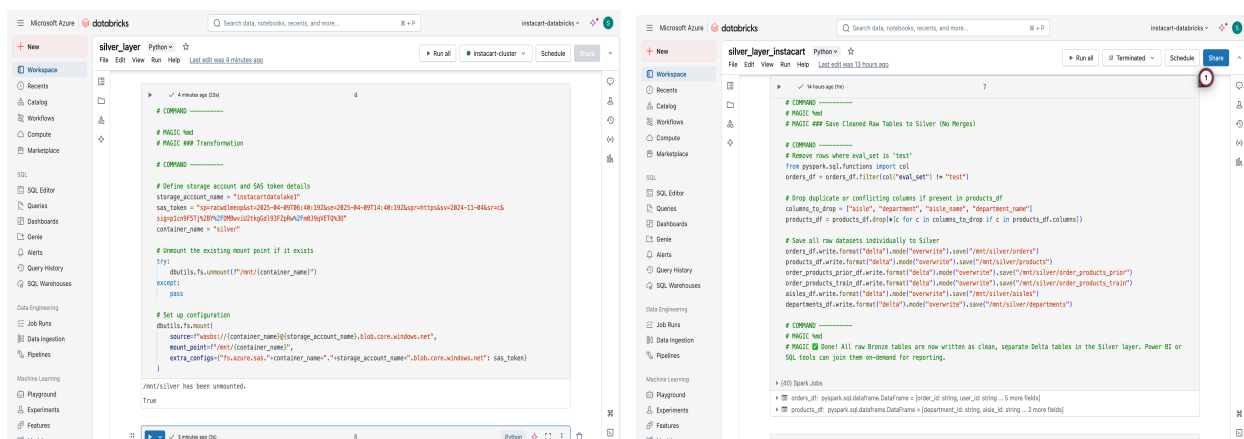
6. Data Transformation Layer

The transformation layer is where raw data is cleaned, filtered, enriched, and shaped for analytics. To perform these operations, an Azure Databricks workspace is created. Databricks provides a collaborative environment for running Apache Spark workloads using notebooks written in Python or SQL.

In this layer:

- Data is read from the Raw (Bronze) layer in ADLS Gen2.
- Using Databricks notebooks, the raw data is cleansed—this might involve removing nulls, fixing data types, standardizing formats, or joining with reference data.
- After transformation, the refined data is saved to the Silver layer of Data Lake Gen2.

This step aligns with the medallion architecture, which separates data into stages (raw → refined → curated) for better governance and traceability. It also ensures the downstream layers get well-structured, analytics-ready data.



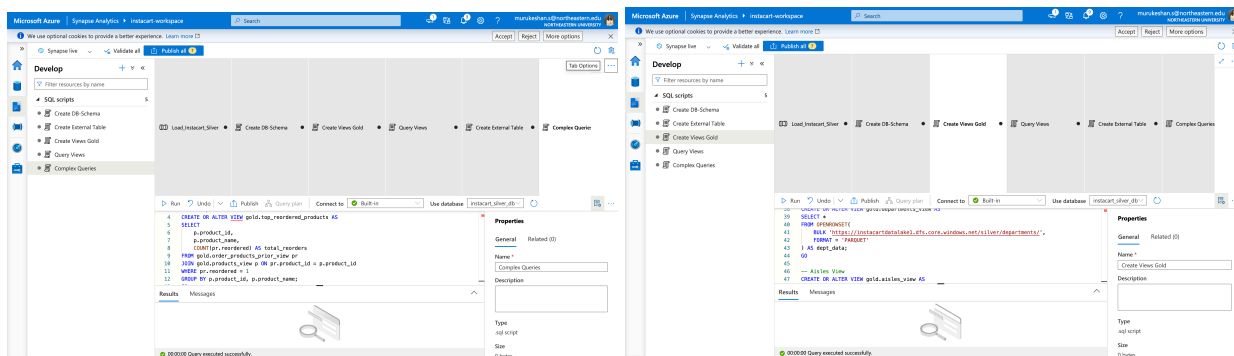
7. Serving Layer

The serving layer is focused on enabling data consumption for analytics and reporting. The first step here is creating an Azure Synapse Analytics workspace. Synapse allows users to query the data directly from the Silver Layer using serverless SQL pools, which is cost-effective and doesn't require provisioning or managing dedicated computation. In Synapse Studio, we created a new SQL script and switched to the default database. Then, created a database-scoped credential using the managed identity or SAS token, followed by defining an external data source pointing to the Silver container. Then, defined a file format for Parquet files and used the OPENROWSET function to query Parquet files from the Silver folder. Finally created SQL views using CREATE OR ALTER VIEW statements, with explicitly defined schemas. We made sure to create both raw views (e.g., orders_view) and analytical views (e.g., top_reordered_products).

Once Synapse is configured and linked with the Data Lake, Power BI is used for visualization. Power BI connects either to Synapse or directly to the Data Lake, allowing for:

- Building interactive dashboards
- Creating report visuals that summarize metrics
- Enabling stakeholders to explore trends and patterns in the data

This layer essentially bridges the gap between processed data and business decision-making, providing accessible insights to end-users through self-service analytics tools.

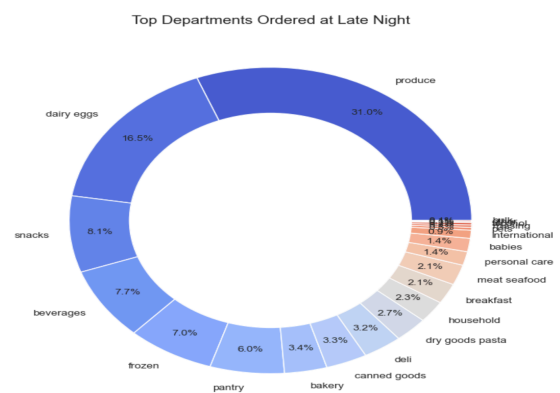
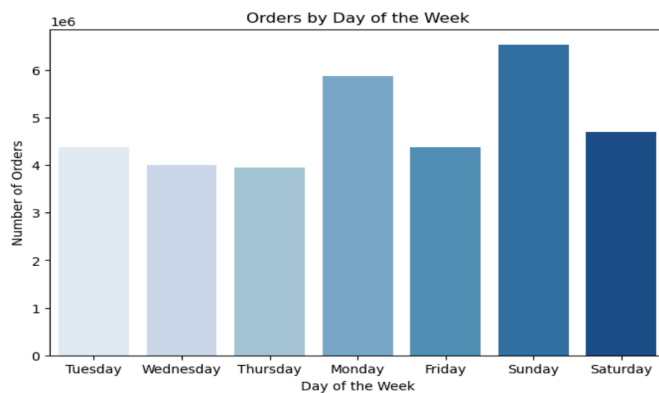


8. Exploratory Data Analysis

Each dataset—aisles, departments, orders, order_products, and products—was scanned for missing and duplicate values as part of the data preparation process. Specifically, null values in the days_since_prior_order column were replaced with 0 to indicate first-time purchases. The products dataset was enriched by merging it with aisles and departments, linking each product to its respective aisle and department. Additionally, the order_products_prior and

order_products_train datasets were concatenated to create a unified product-order relationship. This combined data was further merged with products to include product-level details, and finally, it was merged with orders to create a master transactional dataset containing comprehensive order information.

Exploratory Data Analysis (EDA) on this enriched dataset revealed meaningful customer behavior patterns. Most orders occurred between 9 AM and 5 PM, with 10 AM being the peak hour. Sunday was the most common day for shopping, while weekdays like Wednesday and Thursday saw fewer orders. The days_since_prior_order distribution showed that most customers reordered within a 0 to 30-day window, representing regular monthly or weekly shopping habits. Records with eval_set = "test" were excluded as they are intended for prediction tasks and contained missing values. The analysis also uncovered high-demand departments, providing insight into popular product categories and consumer preferences. These findings form the basis for demand forecasting and personalized recommendation systems.



9. Visualization Layer

The Power BI visualization layer serves as the presentation and analytics interface of the data pipeline. After data is ingested, transformed, and stored in the Silver Layer of Azure Data Lake Gen2, it is made accessible to Power BI for reporting and dashboard creation.

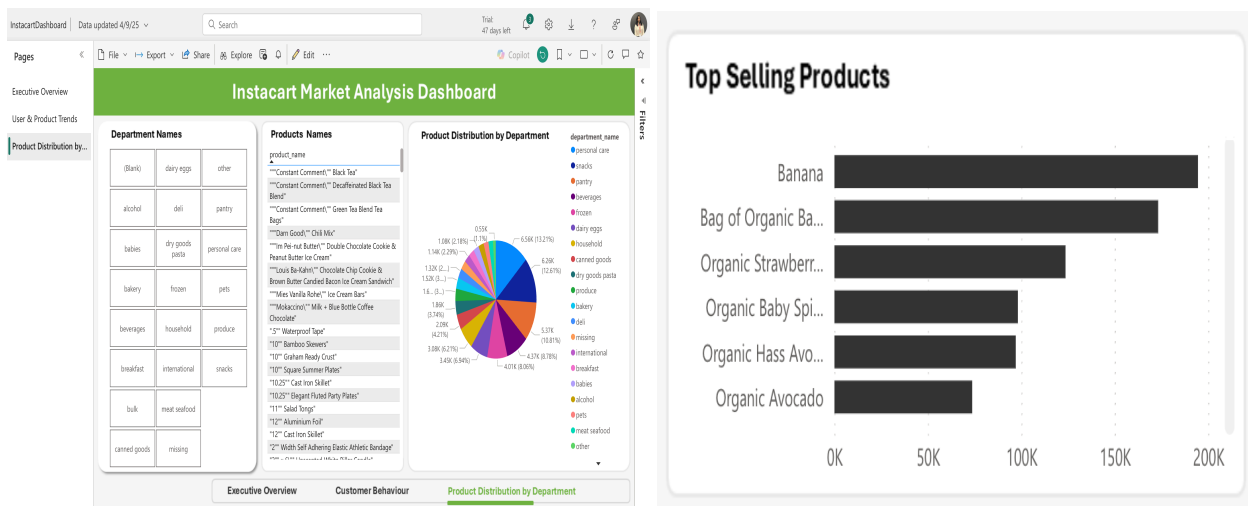
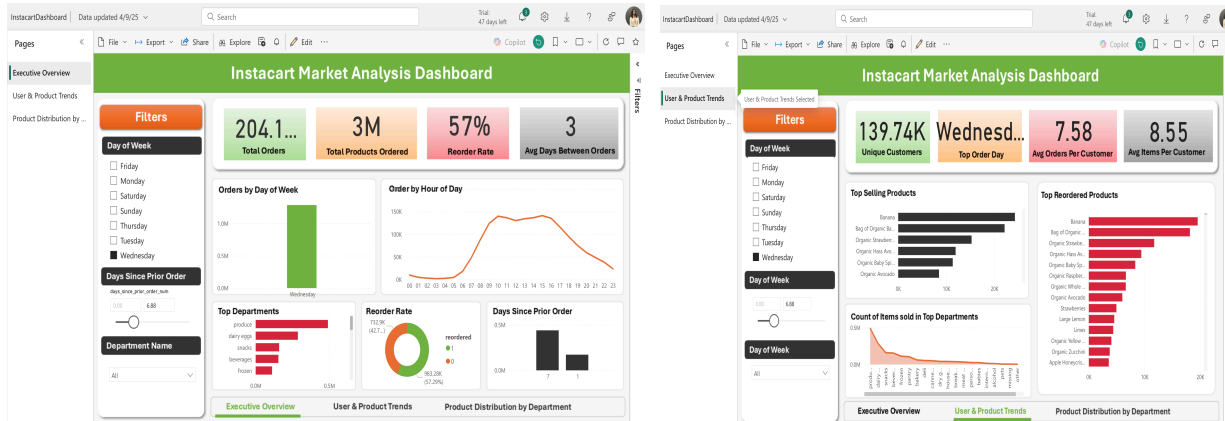
Power BI is connected to the processed dataset, using either Azure Synapse Serverless SQL via Power BI's Azure Data Lake connector. This integration enables seamless access to curated data without exporting it manually. The visuals in Power BI include a range of interactive dashboards, charts, and tables summarizing customer behavior and sales insights from the Instacart dataset. These may cover key KPIs such as:

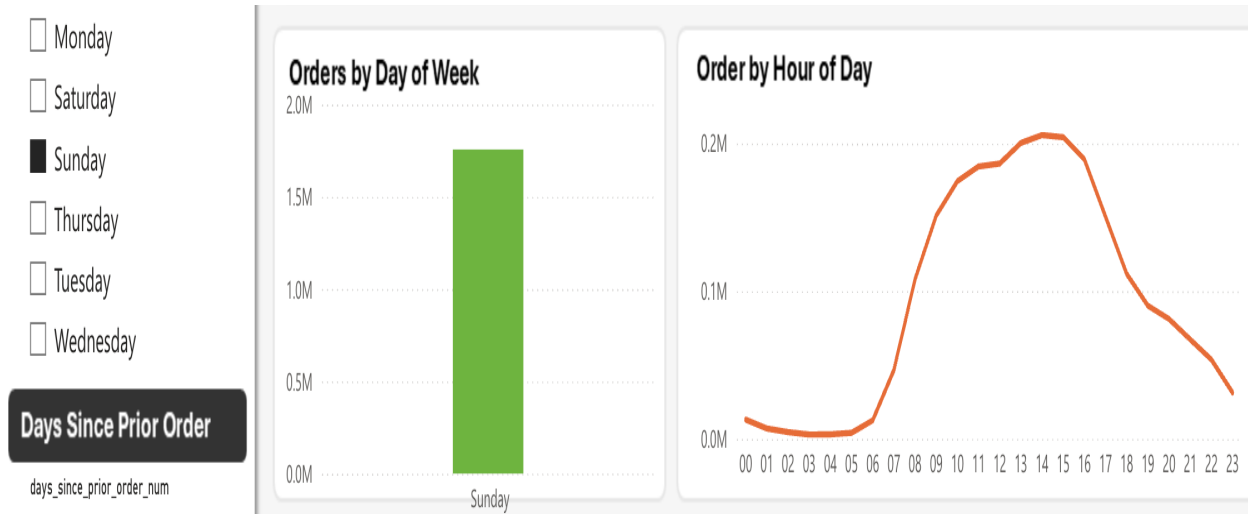
- Most reordered products
- Weekly or hourly ordering trends
- Customer reorder ratios
- Aisles with high product turnover

The goal of this layer is to transform raw insights into actionable business intelligence. Power BI empowers both technical and non-technical users to interact with data, filter reports dynamically, and make data-driven decisions without writing complex queries.

The data analysis from the graph below successfully identifies key consumer purchasing patterns by highlighting the most popular products based on sales volume. It clearly illustrates bananas as the standout product with substantially higher sales compared to others, signifying strong consumer preference. Additionally, the prominence of organic items, including Organic Bananas, Strawberries, Baby Spinach, Hass Avocados, and Avocados, underscores a clear consumer trend towards healthier, organic food options. This insight is valuable for informing strategic business decisions, such as optimizing inventory levels, targeted marketing efforts, and anticipating consumer demand more accurately.

The visualization below highlights bananas as the top reordered product, reflecting high consumer loyalty and frequent repeat purchases. Organic products like strawberries, spinach, and avocados are also significantly reordered, confirming a strong, consistent preference for organic and fresh groceries. Insights from this analysis enable retailers to better predict demand, streamline inventory management, and enhance customer retention by focusing on products with higher reorder frequencies.





10. Key Challenges and Learnings

Challenges:

- Column data types in Parquet sometimes mismatch with the expected SQL schemas.
- SAS tokens expired or failed if not generated with sufficient permissions.
- Synapse Serverless does not support ORDER BY in views, which required us to sort data inside Power BI.

Learnings:

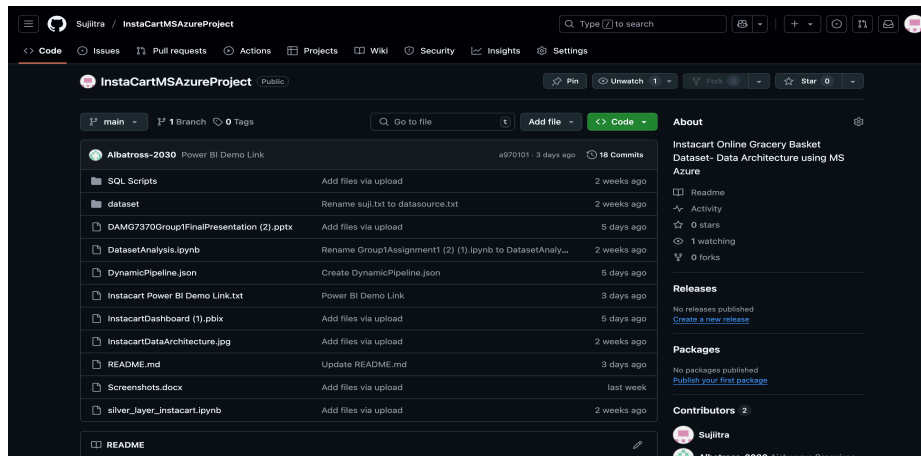
- Medallion architecture enforces structure and reusability.
- PySpark in Databricks is ideal for the distributed processing of large files.
- SQL views are lightweight and performant for data serving.
- Power BI's integration with Synapse enables near real-time dashboards.

11. Artifacts and Resources

- Databricks Notebooks (.ipynb) for transformation logic.
- Synapse SQL Scripts (.sql) for Gold Layer views.
- Power BI Dashboard (.pbix) with custom visuals.
- Architecture Diagram (.jpg).
- Dataset source: <https://www.kaggle.com/datasets/yasserh/instacart-online-grocery-basket-analysis-dataset/>
- GitHub Repository: <https://github.com/Sujiitra/InstaCartMSAzureProject>
- Power BI Demo video: <https://youtu.be/NFxdNhP4VPw?si=5Fu2RGTLiAcuT-Ti>

GitHub Contents:

- silver_layer_instacart.ipynb - Transformation scripts using PySpark in Databricks
- SQL Scripts – Folder with .sql files for Synapse views.
- Dataset - Source dataset link
- DAMG7370Group1FinalPresentation.pptx - Presentation slides
- DatasetAnalysis.ipynb - EDA on the dataset (Assignment1)
- InstacartDataArchitecture.jpg - Architecture flow diagram
- Screenshots.docx - All the screenshots from the project
- DynamicPipeline.json - Azure Data Factory pipeline
- InstacartDashboard.pbix - Power BI file used for reporting
- Power BI Visualizations Demo Video



12. Conclusion

Our project showcases a complete end-to-end data pipeline using Microsoft Azure, following the medallion architecture. Data was ingested from Blob Storage into Azure Data Lake Gen2 using Data Factory, then transformed using Databricks and stored in a refined format. The processed data was queried via Synapse and visualized through Power BI dashboards. This architecture ensures automation, scalability, and effective data-driven insights, making it suitable for real-world analytical applications.

13. References

1. Instacart. (n.d.). *Instacart Market Basket Analysis*. Kaggle. Retrieved from <https://www.kaggle.com/datasets/pspark/instacart-market-basket-analysis>
2. Microsoft. (n.d.). *Power BI documentation*. Microsoft Learn. Retrieved from <https://learn.microsoft.com/en-us/power-bi/>
3. Databricks. (n.d.). *Medallion architecture*. Databricks Glossary. Retrieved from <https://www.databricks.com/glossary/medallion-architecture>